

KolmoX: Structural Preconditioning Architecture for Lossless Heterogeneous Data Compression

Transcending the Limits of Black-Box Entropy Coders via Bit-Exact Domain Decomposition

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Abstract

Conventional general-purpose compression algorithms (e.g., LZMA, Deflate, Zstandard) treat arbitrary inputs as untyped, one-dimensional scalar byte streams. While remarkably effective for textual data exhibiting repeated substring sequences, they systematically underperform on structured numerical, spatial, and floating-point domains—such as continuous trajectories, telemetry, multidimensional scientific floats, raster frames, and audio waveforms—due to high local entropy in lower-order bit planes.

In this paper, we present **KolmoX**, a two-stage hierarchical lossless compression architecture based on structural preconditioning. In the primary stage, a domain-aware engine identifies payload schemas and executes deterministic, bit-exact transforms: columnar channel demuxing, IEEE-754 *byte-plane slicing*, stride autocorrelation, and 2D spatial delta modeling. In the secondary stage, the separated streams are encapsulated into the multi-stream **KMX2** container and encoded via Finite State Entropy (FSE) using Zstandard.

Empirical evaluations conducted exclusively on physical, uncompressed production datasets (including NASA/STScI James Webb Space Telescope sensor arrays, dense CAM toolpaths, industrial telemetry, and 3D geometries) demonstrate solid gains over baseline state-of-the-art compressors across 10 distinct domains. KolmoX achieves up to **+76.54%** net reduction over Zstandard on binary register telemetry (15.70x ratio), **+52.47%** on industrial CSV streams (12.90x ratio), **+36.78%** on CNC G-Code, and **+22.87%** on multi-gigabyte astronomical FITS observations, with streaming throughputs exceeding **920 MB/s**.

Comprehensive Benchmark Results

The following benchmarks report physical empirical measurements obtained on physical hardware (AMD Zen 4 architecture) across uncompressed production datasets. Zero synthetic or zero-fill patterns were used.

Data Domain	Transformation Pipeline	Baseline (Zstd L3)	KolmoX (KMX2)	Gain vs Zstd	Throughput (Comp / Decomp)
Binary Register Packets (.bin)	Stride Autocorr + Demux	3.68x	15.70x	+76.54%	~450 MB/s / ~930 MB/s
Industrial Telemetry (.csv)	Columnar Demux + Quant Delta	6.13x	12.90x	+52.47%	~42 MB/s / ~56 MB/s
CNC G-Code (.gcode)	Columnar Axis Separation	3.46x	5.48x	+36.78%	~35 MB/s / ~42 MB/s
2D Natural Sensor Raster (.bmp)	2D Spatial Delta	2.84x	3.97x	+28.26%	~265 MB/s / ~285 MB/s
Parametric 3D CAD Mesh (.obj)	Ordered Vertex Plane Slicing	3.14x	4.26x	+26.45%	~520 MB/s / ~600 MB/s

Data Domain	Transformation Pipeline	Baseline (Zstd L3)	KolmoX (KMX2)	Gain vs Zstd	Throughput (Comp / Decomp)
Astrophysics FITS (JWST)	2D Modular Delta + Big-Endian Slicing	1.47x	1.90x	+22.87%	~225 MB/s / ~340 MB/s
LiDAR XYZ Point Cloud	Columnar Coordinate Slicing	1.15x	1.44x	+20.00%	~280 MB/s / ~680 MB/s
Dense Vector Buffers (1M Float32)	IEEE-754 Byte-Plane Slicing	1.61x	1.97x	+18.45%	~323 MB/s / ~927 MB/s
Audio PCM 16-bit (.wav)	Stereo Mid/Side Decorrelation	1.45x	1.55x	+6.09%	~135 MB/s / ~260 MB/s
x86 Binary Executable (.exe)	Branch Target Normalizer (BCJ)	1.87x	1.86x	-0.66%	~25 MB/s / ~35 MB/s

Note: Metrics reflect physical bit-exact roundtrips on uncompressed real-world corpora. Compilers, runtimes, and branch predictors favor continuous columnar streams, delivering sub-microsecond decompression latencies (<5 ns per numerical point).

Domain Transform Specifications & Formal Theory

1. Astrophysics FITS & Multidimensional Tensor Modeling

Astronomical instruments (e.g., JWST NIRCam/MIRI, Hubble STIS) produce FITS data cubes formatted in Big-Endian standard byte order with 2880-byte header blocks followed by dense floating-point (>f4) or integer (>i4) matrices.

The `FitsEngine` implements a two-stage preconditioning pipeline:

1. **2D Spatial Modular Delta:** Exploits continuous flux transitions across adjacent focal plane detector pixels:

$$\Delta_{x,y} = (S_{x,y} - S_{x-1,y}) \bmod{2^B}$$

where $S_{x,y}$ is the raw pixel intensity and $B \in \{16, 32\}$ represents bit depth.

2. **Big-Endian Byte-Plane Transposition:** Floating-point matrices isolate the sign bit and exponent (high byte P_3) from the least significant mantissa bits (P_0, P_1). Because physical sky backgrounds exhibit localized thermal equilibrium, P_3 and P_2 collapse into long run-length sequences, allowing entropy coders to process uniform statistical distributions:

$$\mathcal{M}_{i,j} \xrightarrow{\text{slicing}} \{P_0(i), P_1(i), P_2(i), P_3(i)\}$$

2. High-Throughput In-Memory Vector Slicing

In dense vector environments (e.g., neural embeddings, high-frequency telemetry caching), float representations incur severe penalties under LZ77-based match finders due to the chaotic nature of low-order mantissa bits. KolmoX transposes contiguous 32-bit buffers into 4 disjoint orthogonal memory blocks, isolating high-entropy IEEE-754 mantissa noise and enabling memory streaming decompression speeds exceeding **920 MB/s**.

KMX2 Container Specification

The **KMX2** container format implements a fixed 24-byte header supporting multi-stream encapsulation:

```
Header = < Magic(4B), Version(2B), DomainID(1B), Flags(1B), RawSize(8B), AuxLen(8B) >
```